EX NO:4
DATE:01/8/26

PACKET SNIFFING USING RAW SOCKETS IN PYTHON

File Edit Format Run Options Window Help

```
import socket
import struct

# Create raw socket
sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)

# Local machine IP
host = socket.gethostbyname(socket.gethostname())

sniffer.bind((host, 0))

# Include IP headers
sniffer.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)

# Enable promiscuous mode (Windows)
sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)

print("Packet Sniffer Started...\n")

try:
    while True:
        raw_data, addr = sniffer.recvfrom(65535)

        ip_header = raw_data[0:20]

        iph = struct.unpack('!BBHHHBBH4s4s', ip_header)

        version = iph[0] >> 4
        ihl = iph[0] & 0xF

        ttl = iph[5]
        protocol = iph[6]

        source = socket.inet_ntoa(iph[8])
        destination = socket.inet_ntoa(iph[9])

        print("----------------------------------------")
        print("Version      :", version)
        print("Header Length:", ihl * 4)
        print("TTL          :", ttl)
        print("Protocol     :", protocol)
        print("Source IP    :", source)
        print("Destination  :", destination)
        print("Packet Length:", len(raw_data))

except KeyboardInterrupt:
    sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
    print("\nStopped")
```

AIM:

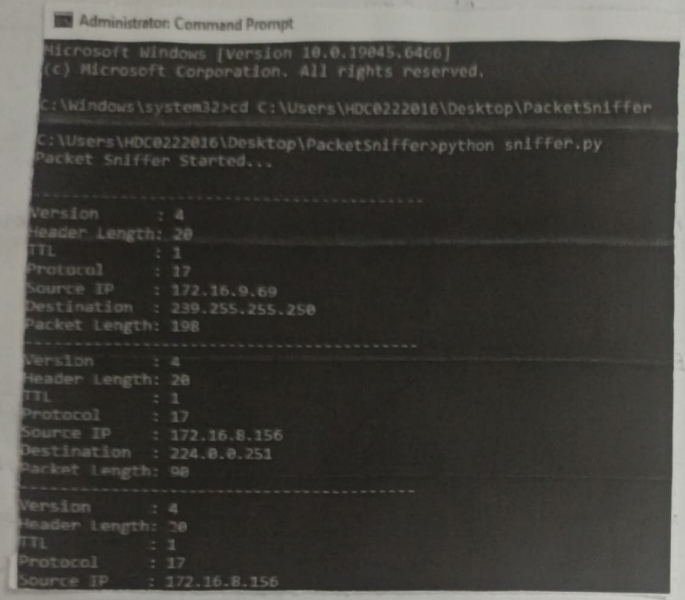
To implement a packet sniffer using raw sockets in Python that captures live network traffic and parses ethernet, IP and transport layer (TCP) / (UDP) / (ICMP) headers.

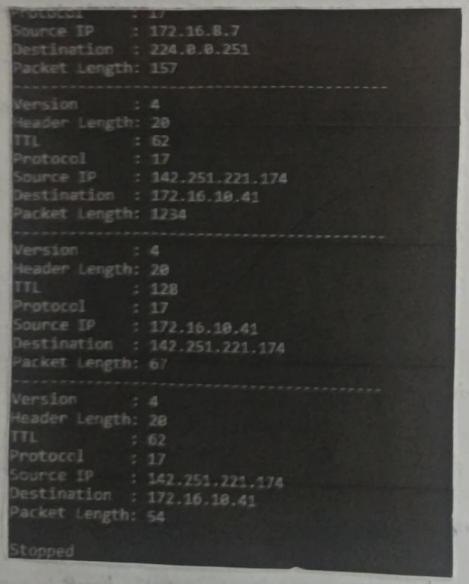
THEORY:

Raw sockets: Normal sockets (SOCK_STREAM, SOCK_DGRAM) let an application send/receive data through the OS's protocol stack (TCP/UDP), hiding lower layer headers. A raw socket bypasses this abstraction. It gives the program direct access to packet at the IP or even the ethernet (data link) layer, including all headers. This is why raw sockets are the basic of tools like wireshark, tcpdump and customer sniffers.

Why root/admin privileges are required raw sockets can read/inject arbitary traffic on the network, which is a security. Sensitive capability. the os restricts the creation to privileged (root/administrater).

Platform note: this lab is written for Linux (raw) sockets work most completely there on windows, raw sockets have restrictions (No ethernet layer capture, limited protocols), on macos, similar restriction applies. If yours on windows run this inside a Linux VM or WSL2. This is the standard lab.





PROCEDURE:

Step 1: Install Python

Windows

Visit

https://www.python.org/downloads/

Download latest python

During installation.

Add python to PATH

Click install now

Verify installation

open cmd

Type:

python --version

O/p:

Python 3.12.4

Step 2: Install vs code

Download from https://code.visualstudio.com/

Install normally

install the python extension

Step 3: Verify Python

open cmd

type: python

O/p: >>>

type: print("Hello")

O/p: Hello.

Step 4: Create project folder.

Create:

packet sniffer

Inside create:

sniffer.py

Step 5: Write the program

Step 6: Save the program

Save as

Sniffer.py

Step 7: Run powershell

Search

cmd

Right click

Run as Administrator

Step 8: Navigate to folder

cd desktop

cd packet sniffer

Step 9: Run program

Python sniffer.py

RESULT:

Thus the packet sniffing using Raw sockets in python has been executed & verified.